

Title

AI-Based Sentiment Analysis System Using Machine Learning and Deep Learning

Objective

The main objective of this project is to develop an AI-based sentiment analysis system that can automatically classify customer or movie reviews as positive or negative. The project uses Natural Language Processing (NLP), Machine Learning, and Deep Learning techniques to analyze textual data and understand public sentiment. The system helps organizations monitor customer opinions, improve services, and make better business decisions.

Introduction

Sentiment Analysis is one of the most important applications of Artificial Intelligence and Natural Language Processing. It is used to identify emotions, opinions, and attitudes expressed in textual data such as social media posts, product reviews, and customer feedback.

In today's digital world, huge amounts of reviews and comments are generated daily on websites and social media platforms. Manually analyzing these reviews is difficult and time-consuming. Therefore, automated sentiment analysis systems are required to process large amounts of text efficiently.

This project focuses on building a sentiment analysis model using Machine Learning algorithms such as Naïve Bayes and Logistic Regression, along with a Deep Learning model called LSTM (Long Short-Term Memory). The model is trained on the IMDB movie review dataset to classify reviews into positive and negative sentiments.

The project demonstrates how AI can understand human language and predict emotions from text data. It also includes preprocessing techniques such as tokenization, stopword removal, stemming, and TF-IDF feature extraction to improve model accuracy.

Dataset Description

The dataset used for this project is the IMDB Movie Review Dataset obtained from Kaggle.

Dataset Link:

[IMDB Dataset on Kaggle](#)

The dataset contains 50,000 movie reviews labeled as either positive or negative. Each review is stored as text data along with its sentiment label.

Dataset Features:

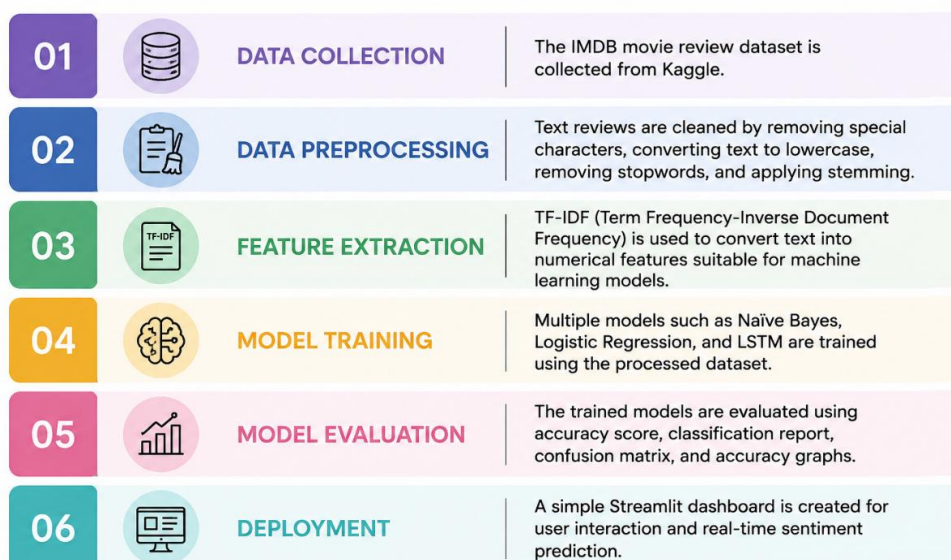
- Review: Textual movie review
- Sentiment: Positive or Negative label

The dataset is balanced, containing an equal number of positive and negative reviews, which helps improve model training and evaluation.

Methodology:

METHODOLOGY

The project follows several stages to build the sentiment analysis system successfully.



Algorithms Used

1. Naïve Bayes

Naïve Bayes is a probabilistic machine learning algorithm commonly used for text classification problems. It predicts sentiment based on word probabilities.

2. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification tasks. It classifies reviews as positive or negative with high accuracy.

3. LSTM (Long Short-Term Memory)

LSTM is a Deep Learning recurrent neural network model designed for sequence and text processing tasks. It remembers long-term dependencies in textual data and improves prediction performance.

Preprocessing

Text preprocessing is an important step in Natural Language Processing. The following preprocessing techniques are used in this project:

- Removal of special characters and punctuation
- Conversion of text into lowercase
- Tokenization of words
- Stopword removal using NLTK
- Stemming using Porter Stemmer
- TF-IDF vectorization for feature extraction

These preprocessing steps help improve model performance and reduce unnecessary information in the dataset.

Model Training

The dataset is divided into training and testing sets using an 80:20 ratio.

- Naïve Bayes and Logistic Regression models are trained using TF-IDF features.
- LSTM model is trained using tokenized and padded text sequences.
- The models learn patterns from the reviews and predict the corresponding sentiment labels.

Training accuracy and validation accuracy are monitored during the training process.

Results

The sentiment analysis system successfully classified movie reviews into positive and negative categories. Multiple machine learning and deep learning models were trained and evaluated.

The Logistic Regression and LSTM models achieved higher accuracy compared to the Naïve Bayes model. The LSTM model provided the best performance because it captures sequential relationships in text data effectively.

The Streamlit dashboard was also able to predict sentiments for custom user reviews successfully.

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Text Preprocessing Completed

TF-IDF Feature Extraction Completed

Training Naïve Bayes Model...
Naïve Bayes Accuracy: 0.825

Training Logistic Regression Model...
Logistic Regression Accuracy: 0.866

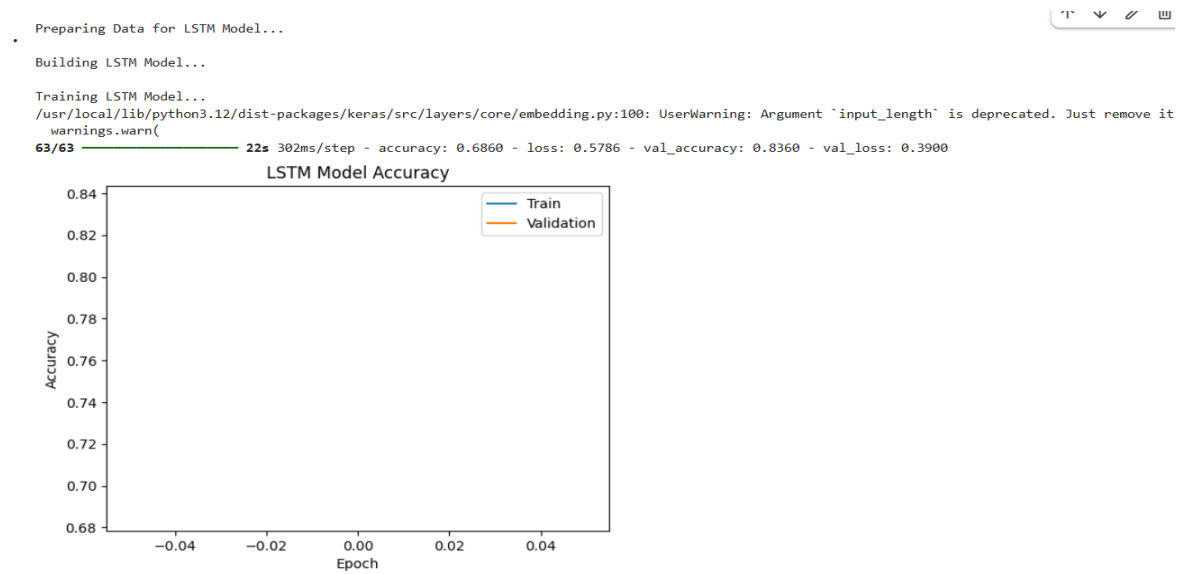
Classification Report:

              precision    recall  f1-score   support

     0           0.89       0.85        0.87         530
     1           0.84       0.88        0.86         470

 accuracy                   0.87         1000
 macro avg              0.87        0.87        0.87         1000
 weighted avg           0.87        0.87        0.87         1000
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Accuracy Comparison:



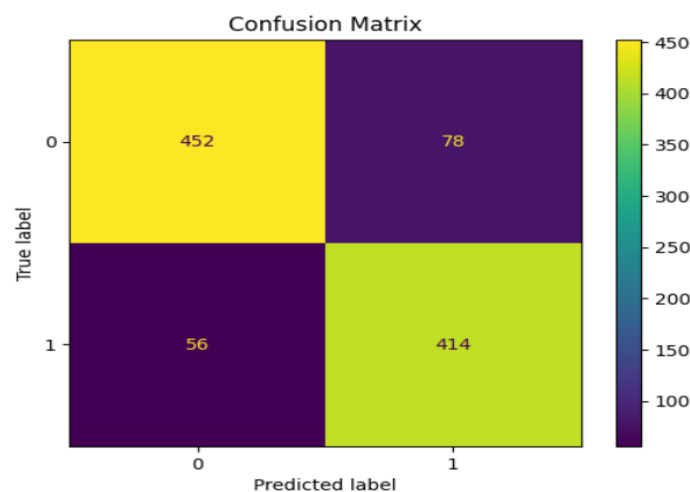
Confusion Matrix

The confusion matrix is used to evaluate the classification performance of the model.

It displays:

- True Positive predictions
- True Negative predictions
- False Positive predictions
- False Negative predictions

The confusion matrix shows that most reviews were classified correctly with minimal prediction errors.



Future Improvements

The following improvements can be implemented in future versions of the project:

1. Add a Neutral sentiment category for better classification.
2. Use advanced transformer models such as BERT for improved accuracy.
3. Perform real-time Twitter sentiment analysis using APIs.
4. Develop multilingual sentiment analysis systems.
5. Deploy the project on cloud platforms for public access.
6. Add graphical dashboards with real-time analytics and visualization.